

RUTVIK PATEL

Senior Data Engineer

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SKILLS

Methodologies: Agile (Scrum, Kanban, Lean), Waterfall, Data Modeling, Data Warehousing, Data Lake Architecture

Programming Languages: Python, Scala, SQL, Java, Bash

Data Management & Analytics: ETL, Data Pipelines, Machine Learning, Data Governance, Data Quality, Data Security, Data Virtualization

Databases & Tools: MySQL, Oracle, Cassandra, Kafka, Spark, Hadoop, Hive, Terraform, Jenkins, Docker, Kubernetes, Ansible, Informatica, Talend, Tableau, Power BI, Looker, Data Profiling, Data Cleansing Tools

Cloud Technologies: AWS (EC2, S3, Lambda, Glue, CloudWatch, IAM, Redshift, ELB, ECS, EKS, EMR), Docker Compose

Frameworks: Django, Flask, Spark Streaming, CI/CD Pipeline

IDEs: PyCharm, PyScripter, Spyder, PyStudio, PyDev

Libraries: Pandas, NumPy, Matplotlib, Boto3, Beautiful Soup, MySQL Connector, New Relic Libraries, Splunk Libraries

Version Control: Git, GitHub

Operating System: Windows, Linux, MacOS

WORK EXPERIENCE

Sr. Data Engineer

Oct 2023 – Present

JPMorgan Chase

- Architected and implemented complex data pipelines leveraging Hadoop, Hive, and Scala to transform raw data into actionable insights.
- Collaborated effectively with cross-functional teams to extract valuable insights, driving data-driven decision making.
- Optimized data processing workflows, resulting in a 40% reduction in processing time.
- Successfully migrated Hadoop on-premise pipelines to AWS, achieving a 30% cost reduction.
- Developed and integrated machine learning models into data pipelines to enhance predictive capabilities.
- Implemented Infrastructure as Code using Terraform to efficiently manage AWS resources.
- Leveraged AWS services (Lambda, Glue, S3) to build scalable and fault-tolerant data processing solutions.
- Provided technical leadership and mentorship to junior team members, fostering a culture of continuous learning.
- Ensured adherence to industry best practices and standards for data management.
- Successfully led the migration of Hadoop pipelines to AWS, resulting in significant cost savings and improved performance.
- Implemented machine learning models that increased predictive accuracy by 40%.
- Optimized data pipelines, leading to a 24% increase in data processing efficiency.

Data Engineer

Aug 2021 – Oct 2023

Capital One

- Architecture and implemented robust RESTful API integrations, meticulously extracting, cleansing, and populating enterprise databases with critical data.
- Engineered a sophisticated near real-time data pipeline utilizing Apache Spark, seamlessly ingesting data from Kafka, executing intricate transformations, and persisting it into Cassandra for actionable insights.
- Developed and orchestrated efficient ETL processes leveraging AWS Glue Studio to seamlessly populate staging databases with diverse datasets from various REST APIs.
- Strategically deployed and managed highly scalable and resilient cloud infrastructure on AWS, employing automation tools such as Python, Boto3, Ansible, and Terraform to optimize resource utilization.
- Containerized and deployed applications on AWS ECS through meticulously constructed CI/CD pipelines utilizing Jenkins, Docker, and GitHub, accelerating delivery cycles.
- Implemented a robust monitoring framework using Splunk and New Relic to proactively identify, diagnose, and resolve system anomalies, ensuring optimal performance and reliability.
- Proactively mitigated security risks by automating vulnerability management processes for SG, White Source, and artifact versions through the implementation of Codeine.
- Demonstrated proficiency in database management by crafting and executing complex MySQL queries using Python-MySQL connector and MySQL DB package.
- Effectively orchestrated Docker containers within Kubernetes clusters, implementing load balancing mechanisms to

optimize resource allocation and application performance.

- Developed and executed Python scripts to automate routine tasks, enhancing operational efficiency and reducing manual intervention.
- Achieved significant performance improvements by optimizing data processing pipelines, resulting in a 33% increase in efficiency.
- Accelerated infrastructure provisioning by 24% through automation, streamlining deployment processes.
- Substantially reduced mean time to resolution by 43% through proactive monitoring and incident response.

Software Developer

May 2020 – Aug 2021

Anthem

- Full-stack development: Proficient in Django and Flask frameworks, architecting robust web applications adhering to MVC principles. Skilled in crafting RESTful APIs for seamless data integration with Oracle and MySQL databases.
- DevOps expertise: Spearheaded CI/CD pipeline implementation using GitLab and Jenkins, optimizing development lifecycle. Integrated New Relic with AWS CloudWatch for advanced monitoring and troubleshooting. Implemented Traefik as a reverse proxy for enhanced system performance. Leveraged AWS Lambda for serverless functions and data management.
- Cloud infrastructure: Managed AWS resources including EC2 instances, ELBs, Auto Scaling groups, and S3 buckets. Implemented load balancing and high availability solutions. Designed and implemented log transformation pipelines from AWS CloudWatch to New Relic for comprehensive data analysis.
- Data management: Built and maintained efficient data pipelines, including data extraction, transformation, and loading processes. Utilized Splunk for log analysis and troubleshooting.

Python Developer

Oct 2017 – May 2019

Prompt SoftTech

- Developed and meticulously documented Waterfall business process models to provide a comprehensive blueprint for current and future operational procedures.
- Leveraged the capabilities of Python libraries such as Pandas, SciPy, Matplotlib, and NumPy to conduct in-depth analysis of complex datasets, extracting valuable insights and informing strategic decision-making.
- Constructed intricate SQL queries to rigorously test the database system's functionality, ensuring data integrity and optimizing query performance.
- Designed and implemented secure RESTful web services utilizing the Flask framework and robust authentication mechanisms provided by Flask-HTTP Auth and HTTPS to protect sensitive data and maintain system integrity.
- Established and maintained a robust DevOps pipeline, incorporating Docker, Jenkins, and Git, to streamline the development, testing, and deployment of microservices, ensuring efficient and continuous delivery of software.
- Developed highly optimized Django ORM queries to significantly reduce database load and improve overall system performance by preloading data and minimizing redundant database interactions.
- Created and maintained intuitive dashboard control panels using Django, HTML, CSS, JavaScript, Bootstrap, and jQuery to provide stakeholders with real-time insights and actionable information.
- Implemented effective version control practices using GitHub to track code changes and maintain code integrity, while also producing detailed release notes to document software updates and enhancements.

EDUCATION

NJIT

May 2021

Master in Information Systems

LJIET

May 2019

Bachelors in Computer Engineering

CERTIFICATION

- AWS Professional Certified
- Terraform Certified